

Canonical signed common terminal, trace-bracket defect, and affine-ray boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

The purpose of this result is to discover what the registered action actually permits, not to force BCC or any other spatial phase. Every statement below is independent of whether a later minimizer is modulated, uniform, disordered, absent, or outside the present scalar working branch.

R-146 supplied a new fixed-cutoff Boue–Dupuis chart whose source covariance increments are proportional to the complete terminal covariance. It left a fail-closed sequence: construct one complete terminal feature, build its common-terminal tower, and then decide whether the old trace, future, forest, and low owners may be identified with that tower.

R-130 already established the generic legal principle of recombining first and rerunning Doob for its fixed Ξ terminal. This note does not reclaim that principle. Its new content is the signed R-141/R-142 terminalization in the R-146 canonical chart, the polarized trace–bracket defect and transfer criterion, the sharper centred counterfixtures, and the exact production Pauli coefficient slices.

R-147 conclusion. For the actual R-141/R-142 trace–current terminal feature at fixed cutoff and the declared canonical source filtration, there is a unique common-terminal signed Doob tower. Its discrete and continuous signed telescopes give the exact relative aggregate terminal-feature value while retaining every mixed endpoint correlation.

This new aggregate coordinate does not by itself establish transport of the old R-104/R-063/R-125 owners. A representation-universal transport is equivalent to intertwining the entire conditional-expectation nest. For the actual terminal alone, proposed rows must instead pass the tower condition (4.6), the registered incidence hypotheses, and the polarized predictable trace–bracket matching conditions of Section 5.

The exact production Pauli $P + L$ pair is strictly favourable on every predictable affine collinear Gaussian line, with uniform curvature

$$H'' \geq \frac{2223}{25000P}.$$

That geometry is sharp: an explicit active–spectator direction has negative curvature and hence a positive small-noise trace–current defect. Therefore closure based solely on global convexity of this $P + L$ coefficient pair cannot close the scalar sea. The complete non-collinear future/forest/balanced/low signed residual remains the live production target. T-050, A13, Nelson, every phase question, and Sector A remain open at T4.

2. The actual terminal feature and auxiliary-variable firewall

Write the probability space as

$$\Omega = \Omega_{\text{src}} \times \Omega_{\text{aux}}. \quad (2.1)$$

The source factor carries the canonical Brownian filtration $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_N$. The auxiliary factor contains the independent target-heat dummy, the shared covariance probe, and only those replica coordinates already declared auxiliary by the registered owner. Hilbertize them as

$$\mathcal{K} = L^2(\Omega_{\text{aux}}; \mathcal{H}), \quad (2.2)$$

with finite orthogonal direct sums when needed. Source conditional expectation is the Bochner projection

$$P_b = \mathbb{E}_{\text{src}}[\cdot \mid \mathcal{F}_b] \otimes \text{I}_{\mathcal{K}}. \quad (2.3)$$

Every deterministic bounded operator S on $\mathcal{K}$ commutes with P_b :

$$P_b S = S P_b. \quad (2.4)$$

This is the load-bearing firewall. Averaging the shared heat or probe before forming the feature would change the mixed trace Gram.

For each endpoint $q \in \{h, 0\}$, use the actual R-142 shared-probe trace feature U_q , the complete current J_q , and the same legal source projection $P_f = \mathbb{E}[\cdot \mid \mathcal{F}_f]$, with f included in the nested reveal grid. Put

$$\Phi_q = P_f J_q, \quad R_q = (\text{I} - P_f) J_q, \quad (2.5)$$

$$F_q = (U_q, \Phi_q), \quad S_2 = \text{diag}(+\text{I}, -\text{I}), \quad (2.6)$$

$$Z_q = (U_q, J_q, R_q), \quad S_3 = \text{diag}(+\text{I}, -\text{I}, +\text{I}). \quad (2.7)$$

The word “actual” means that both endpoints use the same covariance square root, probe, heat coordinate, output projection, and future coupling. A vector with the same diagonal norm is not an admissible substitute.

Theorem 2.1 (polarized three-to-two feature compression)

For $p, q \in \{h, 0\}$, orthogonality of the range and kernel of the common source projection P_f in the full source–auxiliary L^2 space gives

$$\boxed{\mathbb{E} \text{Re} \langle Z_q, S_3 Z_p \rangle = \mathbb{E} \text{Re} \langle F_q, S_2 F_p \rangle = \mathbb{E} \text{Re} \langle U_q, U_p \rangle - \mathbb{E} \text{Re} \langle \Phi_q, \Phi_p \rangle.} \quad (2.8)$$

Consequently the complete R-141/R-142 owner has orientation

$$\mathcal{P}_{\text{comp}}(q) = -\frac{1}{2} \mathbb{E} \langle F_q, S_2 F_q \rangle = -\frac{1}{2} \mathbb{E} \langle Z_q, S_3 Z_q \rangle. \quad (2.9)$$

Proof. Write $J_q = \Phi_q + R_q$. The two mixed terms vanish in the full source–auxiliary L^2 inner product, so $\mathbb{E} \langle J_q, J_p \rangle = \mathbb{E} \langle \Phi_q, \Phi_p \rangle + \mathbb{E} \langle R_q, R_p \rangle$. Substitution in the S_3 form proves (2.8). Equation (2.9) is the exact R-141/R-142 action convention. $\square$

If P_f is one fixed member of the same nested source filtration, then $P_b P_f = P_f P_b$. Before f , $dJ = d\Phi$ and $dR = 0$; after f , $d\Phi = 0$ and $dR = dJ$. Hence (2.8) also holds increment by increment. No old root or forest owner has been invoked.

3. Exact common-terminal signed Doob theorem

Set

$$F^\Sigma = F_h + F_0, \quad F^\Delta = F_h - F_0, \quad (3.1)$$

and define the unique common-terminal martingales

$$M_b^a = P_b F^a, \quad d_b^a = M_b^a - M_{b-1}^a, \quad a \in \{\Sigma, \Delta\}. \quad (3.2)$$

Theorem 3.1 (discrete signed terminal telescope)

At every fixed cutoff and finite canonical source chart,

$$\begin{aligned} & \mathbb{E}\{\langle F_h, S_2 F_h \rangle - \langle F_0, S_2 F_0 \rangle\} \\ &= \mathbb{E} \operatorname{Re} \langle M_0^\Sigma, S_2 M_0^\Delta \rangle + \sum_{b=1}^N \mathbb{E} \operatorname{Re} \langle d_b^\Sigma, S_2 d_b^\Delta \rangle \end{aligned} \quad (3.3)$$

$$\Delta \mathcal{P}_{\text{comp}} = -\frac{1}{2} \left[\mathbb{E} \operatorname{Re} \langle M_0^\Sigma, S_2 M_0^\Delta \rangle + \sum_b \mathbb{E} \operatorname{Re} \langle d_b^\Sigma, S_2 d_b^\Delta \rangle \right]. \quad (3.4)$$

The same formulas hold for (Z, S_3) .

Proof. Polarization gives $\langle F_h, S F_h \rangle - \langle F_0, S F_0 \rangle = \operatorname{Re} \langle F^\Sigma, S F^\Delta \rangle$. Expand each terminal martingale as $M_0 + \sum_b d_b$. When $b < c$, d_b^Σ is $\mathcal{F}_{c-1}$ -measurable and $\mathbb{E}[S d_c^\Delta | \mathcal{F}_{c-1}] = 0$ by (2.4). Every off-diagonal reveal pair vanishes in expectation, which proves (3.3). Equation (3.4) follows from (2.9). $\square$

Define the conditional signed future remainder

$$V_b^S = \mathbb{E} \left[\operatorname{Re} \langle F^\Sigma - M_b^\Sigma, S (F^\Delta - M_b^\Delta) \rangle \middle| \mathcal{F}_b \right]. \quad (3.5)$$

The same calculation gives

$$V_{b-1}^S = \mathbb{E}[V_b^S | \mathcal{F}_{b-1}] + \mathbb{E}[\operatorname{Re} \langle d_b^\Sigma, S d_b^\Delta \rangle | \mathcal{F}_{b-1}], \quad V_N^S = 0. \quad (3.6)$$

Because S is indefinite, neither (3.3) nor (3.6) has a sign. Their value is exact aggregate terminal-feature accounting without deleting favourable current correlations.

Theorem 3.2 (continuous Brownian form)

Assume that $\mathcal{K}$ is separable and that the completed, right-continuous canonical Brownian source filtration has the martingale-representation property. For an $L^2(\mathcal{K})$ -valued terminal feature, Hilbert-space martingale representation then gives predictable Hilbert–Schmidt integrands H_s^a with

$$M_t^a = M_0^a + \int_0^t H_s^a \, dB_s. \quad (3.7)$$

Define

$$\Delta Q_S = \operatorname{Re} \langle F_h, S F_h \rangle - \operatorname{Re} \langle F_0, S F_0 \rangle.$$

Therefore

$$\mathbb{E} \Delta Q_S = \mathbb{E} \operatorname{Re} \langle M_0^\Sigma, S M_0^\Delta \rangle + \mathbb{E} \int_0^T \operatorname{Re} \operatorname{Tr}[(H_s^\Sigma)^* S H_s^\Delta] \, ds. \quad (3.8)$$

Moreover

$$V_t^S = \mathbb{E} \left[\int_t^T \operatorname{Re} \operatorname{Tr}[(H_s^\Sigma)^* S H_s^\Delta] \, ds \middle| \mathcal{F}_t \right]. \quad (3.9)$$

Martingale representation is sufficient for (3.8). The stronger Clark–Ocone identification $H_s = \mathbb{E}[D_s F \mid \mathcal{F}_s]$ requires $F \in \mathbb{D}^{1,2}$. For the defined shifted terminal

$$X_T = \int_0^T K_{\text{can}}(t)(dB_t + h_t dt),$$

with deterministic K_{can} and adapted, square-integrable, Malliavin-differentiable h ,

$$D_s X_T = K_{\text{can}}(s) + \int_s^T K_{\text{can}}(t) D_s h_t dt. \quad (3.10)$$

Deleting the integral in (3.10) is precisely the future-feedback error that the R-136 raw sequential owner forbids.

4. Exact filtration-nest transfer criterion

Theorem 3.1 creates new canonical aggregate owners. It does not say that an old root family is that tower.

Theorem 4.1 (conditional-expectation nest intertwining)

Let $W : L^2(\Omega_1; \mathcal{K}) \rightarrow L^2(\Omega_2; \mathcal{K})$ be a unitary induced by a probability-algebra isomorphism and commuting with S . Let $P_b^{(i)}$ be the conditional projections of two filtrations. Then

$$W L^2(\mathcal{F}_b^{(1)}; \mathcal{K}) = L^2(\mathcal{F}_b^{(2)}; \mathcal{K}) \quad \text{for all } b. \quad (4.1)$$

implies

$$W P_b^{(1)} = P_b^{(2)} W. \quad (4.2)$$

Thus $W F_*^{(1)} = F_*^{(2)}$ transports every martingale row and signed bracket. Conversely, transport of conditional projections for every terminal L^2 feature forces (4.1).

Proof. Conditional expectation is the orthogonal projection onto the declared closed subspace. A unitary carrying one range onto the other conjugates the unique orthogonal projections. The converse follows from equality of their ranges. $\square$

For Gaussian linear source processes, a causal physical intertwiner with output unitary V and time change θ necessarily satisfies the whole prefix-covariance identity

$$C_2(t) = V C_1(\theta(t)) V^*, \quad C_i(t) = \int_0^t K_i(s) K_i(s)^* ds. \quad (4.3)$$

Terminal covariance equality is only (4.3) at $t = T$.

Exact rank counterfixture

On $\mathbb{R}^2$, put

$$K_{\text{old}}(t) = \begin{cases} \sqrt{2} P_1, & 0 < t \leq 1/2, \\ \sqrt{2} P_2, & 1/2 < t \leq 1, \end{cases} \quad K_{\text{can}}(t) = I. \quad (4.4)$$

Both terminal covariances equal I , but

$$C_{\text{old}}(1/2) = P_1, \quad C_{\text{can}}(1/2) = \frac{1}{2} I. \quad (4.5)$$

Their ranks are one and two at equal clock time. The obstruction survives an arbitrary time change: the eigenvalues of $C_{\text{old}}(s)$ are

$$(2s, 0) \quad (0 \leq s \leq 1/2), \quad (1, 2s - 1) \quad (1/2 < s \leq 1),$$

so no old time has the canonical half-time spectrum $(1/2, 1/2)$. Thus no causal Gaussian intertwiner which also maps the declared physical process and terminal synthesis can satisfy (4.3). The bare Brownian filtrations remain abstractly isomorphic. This sharpens, rather than contradicts, the R-104 rule: only subdivisions preserving the original filtration, reveal order, and admissible controls transport its recombined endpoint identity.

For a proposed old row family $\widehat{M}_b$, the elementary fail-closed criterion is therefore

$$\widehat{M}_b \in L^2(\mathcal{F}_b), \quad \widehat{M}_N = F_*, \quad \mathbb{E}[\widehat{M}_b \mid \mathcal{F}_{b-1}] = \widehat{M}_{b-1}. \quad (4.6)$$

Conditions (4.6) are necessary and sufficient for $\widehat{M}_b = \mathbb{E}[F_* \mid \mathcal{F}_b]$. The registered family $\Phi_1 = \xi_1$, $\Phi_2 = 2\xi_1 + \xi_2$ fails (4.6).

5. Polarized trace–bracket defect and the R-063 boundary

Fix an owner $o = (k, \mathcal{C})$, and use one shared covariance probe at endpoints p, q . Throughout this section assume the full R-125 incidence: one common heat, the same once-owned covariance, same-root visit recombination, an exhaustive orthogonal output family (or an explicitly retained complement leakage), and prior identification of the constructed partial-Wick coordinate with the registered adapted R-063 forest. Write the R-125 future covariance and primitive trace as

$$v_o[p, q] = \mathbb{E}_f \langle J_{p,o} - \Phi_{p,o}, J_{q,o} - \Phi_{q,o} \rangle, \quad (5.1)$$

$$\tau_o[p, q] = \mathbb{E}_{f, G'} \langle P_{\mathcal{C}} C_6(W_p) \partial G'_o, P_{\mathcal{C}} C_6(W_q) \partial G'_o \rangle. \quad (5.2)$$

Because W_u contains the current root, this primitive trace is generally $\mathcal{F}_o$ -measurable rather than predictable at $\mathcal{F}_{o-}$. Define

$$\bar{\tau}_o[p, q] = \mathbb{E}[\tau_o[p, q] \mid \mathcal{F}_{o-}]. \quad (5.2a)$$

Polarization of R-125 equations (3.3) and (4.5) gives the exact adapted forest coordinate

$$f_o[p, q] = v_o[p, q] + \langle \Phi_{p,o}, \Phi_{q,o} \rangle - \tau_o[p, q], \quad (5.3)$$

and therefore

$$r_o[p, q] := v_o[p, q] - f_o[p, q] = \tau_o[p, q] - \langle \Phi_{p,o}, \Phi_{q,o} \rangle. \quad (5.4)$$

Suppose the candidate mean rows have passed (4.6). Put

$$b_{u,o} = \mathbb{E}[\Phi_{u,o} \mid \mathcal{F}_{o-}], \quad Y_{u,o} = \Phi_{u,o} - b_{u,o} = P_{\mathcal{C}} dM_{u,k}, \quad (5.5)$$

$$\beta_o[p, q] = \mathbb{E}[\langle P_{\mathcal{C}} dM_{p,k}, P_{\mathcal{C}} dM_{q,k} \rangle \mid \mathcal{F}_{o-}], \quad \delta_o[p, q] = \bar{\tau}_o[p, q] - \beta_o[p, q]. \quad (5.6)$$

Theorem 5.1 (matching inside the registered R-125 owner system)

After conditional expectation at the previous reveal,

$$\boxed{\mathbb{E}[r_o[p, q] \mid \mathcal{F}_{o-}] = \delta_o[p, q] - \langle b_{p,o}, b_{q,o} \rangle.} \quad (5.7)$$

Under the preceding R-125/R-063 incidence hypotheses, the conditionally averaged owner residual equals the specific defect-free predictable Gram $-\langle b_{p,o}, b_{q,o} \rangle$ for every polarized endpoint pair if and only if

$$\boxed{\bar{\tau}_o[p, q] = \beta_o[p, q] \quad \text{for every } p, q.} \quad (5.8)$$

Equivalently,

$$\mathbb{E}[f_o[p, q] \mid \mathcal{F}_{o-}] = \mathbb{E}[v_o[p, q] \mid \mathcal{F}_{o-}] + \langle b_{p,o}, b_{q,o} \rangle. \quad (5.9)$$

For only the exhaustive low-plus-root aggregate, the weaker necessary and sufficient condition for reproducing the exact aggregate mean-Gram identity within that same registered owner system is $\sum_o \mathbb{E} \delta_o[p, q] = 0$ for every p, q . Diagonal equality is insufficient for a secant or Hessian; the shared-probe bilinear equality is load bearing. Tower and bracket matching alone are not a standalone old-owner transport theorem: the registered forest identification, heat, covariance incidence, recombination, and complement conditions stated above remain additional proof obligations.

The conditioning in (5.2a) is load bearing. With trivial past, let $\xi = \pm 1$ equiprobably, take $p = q$, $\Phi = \xi$, $b = 0$, $\beta = 1$, and the nonnegative current-root trace $\tau = 1 + \xi$. Then $\tau \neq \beta$ pointwise, while $\bar{\tau} = \mathbb{E}\tau = 1 = \beta$, and

$$\mathbb{E}[r \mid \mathcal{F}_{\text{past}}] = \mathbb{E}(\tau - \Phi^2) = 0 = -b^2. \quad (5.9a)$$

Thus raw pointwise matching is not necessary; predictable matching is.

The R-063 forest is not a quadratic variation or a positive Malliavin Dirichlet energy. It is the same-regulator signed Wick–Taylor reconstruction. At finite cutoff its adapted coordinate contains

$$F : Q = F \diamond Q + Y_\lambda D_\mu F + Y_\mu D_\lambda F - D_\lambda D_\mu F. \quad (5.10)$$

No registered theorem represents (5.10) as $\langle QU_p, QU_q \rangle$ for one orthogonal projection Q . Gaussian Poincare therefore cannot silently replace the forest.

Even if (5.8) holds at both endpoints, put $\delta b_o = b_{h,o} - b_{0,o}$. The relative expected residual is

$$\Delta R^{\text{full}} = - \sum_o \mathbb{E}\{2 \operatorname{Re}\langle b_{0,o}, \delta b_o \rangle + \|\delta b_o\|^2\}. \quad (5.11)$$

It is indefinite unless $b_0 = 0$ or a separately proved complete-low identity retains the baseline cross. The scaled R-125 two-root fixture has $R(a) = -a^2/4$: the changes $2 \rightarrow 1$ and $1 \rightarrow 2$ are $+3/4$ and $-3/4$.

6. Common terminal does not imply scalar sign

Let $t > 0$, and let g, ζ be independent standard Gaussians and

$$A_t(g) = e^{-tg^2/2}, \quad U = A_t(g)\zeta, \quad \Phi = A_t(g)g. \quad (6.1)$$

This is a centred one-step common-terminal current: $\mathbb{E}\Phi = 0$. Nevertheless

$$\mathbb{E}|U|^2 = (1 + 2t)^{-1/2}, \quad \mathbb{E}|\Phi|^2 = (1 + 2t)^{-3/2}, \quad (6.2)$$

so

$$\boxed{\mathbb{E}(|U|^2 - |\Phi|^2) = \frac{2t}{(1 + 2t)^{3/2}} > 0, \quad \frac{\mathbb{E}(|U|^2 - |\Phi|^2)}{\mathbb{E}|U|^2} = \frac{2t}{1 + 2t} \rightarrow 1.} \quad (6.3)$$

Thus common terminal, zero predictable mean, and equal probe covariance do not imply trace–bracket matching. This is a method no-go, not a production or phase counterexample.

Endpoint-law equality also does not preserve owners. The charts $G_A = \xi_1$ and $G_B = (\xi_1 + \xi_2)/\sqrt{2}$ both have law $N(0, 1)$. Their two reveal energies are

$$(1, 0) \quad \text{and} \quad (1/2, 1/2) \quad (6.4)$$

for G , and

$$(2, 0) \quad \text{and} \quad (1/2, 3/2) \quad (6.5)$$

for $G^2 - 1$. Only the totals agree. Equations (4.5), (6.4), and (6.5) show exactly why R-146’s preserved endpoint Boue–Dupuis value alone cannot certify preservation of the old owners.

7. Exact production affine collinear sign

The generic no-go in Section 6 does not decide the registered Pauli coefficient. On a normalized scalar collinear block put

$$\alpha = \frac{5}{9}, \quad a = 1 - \alpha = \frac{4}{9}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \quad (7.1)$$

The exact P - and L -row coefficient energies are

$$H_P(x) = 4c_0x^2, \quad H_L(x) = 4c_1x^2 \left(\frac{ax^2 + e}{x^2 + e} \right)^2, \quad H = H_P + H_L. \quad (7.2)$$

Theorem 7.1 (uniform affine-ray production sign)

For every $e > 0$ and production normalization $P > 0$,

$$H''(x) = \frac{3}{1000P} \frac{113e^4 - 88e^3x^2 + 243e^2x^4 + 192ex^6 + 48x^8}{(e + x^2)^4} \geq \frac{2223}{25000P}. \quad (7.3)$$

Consequently, for every $m \in L^2(\mathcal{F}_{\text{past}})$, every $\sigma > 0$, and a fresh standard Gaussian g independent of that past,

$$\mathbb{E}\{\sigma^2 H(m + \sigma g)(1 - g^2)\} = -\sigma^4 \mathbb{E}H''(m + \sigma g) \leq -\frac{2223}{25000P}\sigma^4 < 0. \quad (7.4)$$

Proof. Direct differentiation gives (7.3). With $y = x^2/e$, the rational numerator ratio is

$$R(y) = \frac{113 - 88y + 243y^2 + 192y^3 + 48y^4}{(1 + y)^4}, \quad (7.5)$$

and

$$R'(y) = \frac{30(y + 9)(3y - 2)}{(1 + y)^5}. \quad (7.6)$$

Its unique nonnegative minimum occurs at $y = 2/3$, where $R = 741/25$. This proves (7.3). Gaussian integration by parts gives $\mathbb{E}[H(m + \sigma g)(1 - g^2)] = -\sigma^2 \mathbb{E}H''(m + \sigma g)$, which proves (7.4). $\square$

At zero background the rows are separately favourable:

$$\mathbb{E}H_P(g)(1 - g^2) = -\frac{12}{125P}, \quad \mathbb{E}H_L(g)(1 - g^2) \leq -\frac{6}{125P}, \quad (7.7)$$

so their pair is at most $-18/(125P)$. The P row is load bearing for translated lines. The normalized L -row curvature at $x^2 = e/2$ is exactly

$$-\frac{56}{2187}. \quad (7.8)$$

For completeness, the separate L -row inequality in (7.7) is not a curvature consequence. Put $u = g^2$, $b = 1 - a$, and

$$w_e(u) = \frac{u\{2au + (1 + a)e\}}{(u + e)^2}.$$

The pointwise decomposition

$$u \left(\frac{au + e}{u + e} \right)^2 = a^2u + be w_e(u)$$

and the Gaussian moments $\mathbb{E}u = 1$, $\mathbb{E}u^2 = 3$ give

$$\mathbb{E} \left[u \left(\frac{au + e}{u + e} \right)^2 (1 - u) \right] = -2a^2 + be \mathbb{E}[w_e(u)(1 - u)]. \quad (7.7a)$$

For $a = 4/9$,

$$w'_e(u) = \frac{e\{(3a-1)u + (1+a)e\}}{(u+e)^3} \geq 0.$$

Since u and $w_e(u)$ are increasing, $\mathbb{E}[w_e(u)(1-u)] = -\text{Cov}(w_e(u), u) \leq 0$. Multiplication by $4c_1$ gives $-8c_1a^2 = -6/(125P)$, proving the second inequality in (7.7).

Theorem 7.1 is a genuine production-coefficient same-root advance. It does not cover non-collinear internal directions, multiple roots, root-dependent backgrounds, active wedge rows, spatial output projections, or the complete future/forest/low owner.

8. Sharp non-collinear production boundary

The favourable ray theorem cannot be globalized by convexity. Take one real active coordinate and one real spectator, set the base to (R, R) , and perturb in direction $(1, -1)$:

$$x(q) = R + q, \quad y(q) = R - q, \quad d(q) = x(q)^2 + y(q)^2 + e. \quad (8.1)$$

The wedge rows vanish. With $t(q) = x(q)^2/d(q)$, the exact production pair along this line is

$$f(q) = 4c_0x(q)^2 + 4c_1\{x(q) - \alpha t(q)[x(q) - y(q)]\}^2. \quad (8.2)$$

Direct differentiation gives

$$f''(0) = \frac{3\{-528R^4 - 88R^2e + 113e^2\}}{1000P(2R^2 + e)^2}. \quad (8.3)$$

It is negative whenever

$$\frac{R^2}{e} > -\frac{1}{12} + \frac{5\sqrt{154}}{132} = 0.3867300623\dots, \quad (8.4)$$

and, for fixed $R \neq 0$, in the formal zero-floor limit $e \downarrow 0$,

$$f''(0) = -\frac{99}{250P}. \quad (8.5)$$

For sufficiently small same-root noise $q = \sigma g$, Stein's identity turns negative curvature into a positive trace-current defect:

$$\sigma^2\mathbb{E}\{f(\sigma g)(1 - g^2)\} = -\sigma^4\mathbb{E}f''(\sigma g) > 0. \quad (8.6)$$

This is an exact production-coefficient counterdirection to global multivariate convexity. It is not a counterexample to the complete action: R-063 forest, future variance, balanced cross terms, returned low, spatial owners, and terminal sextic are absent from (8.2). Those companions must now be tested on this direction rather than assumed favourable or discarded.

9. Proof-route map and revised next target

The successful and failed routes are now sharply separated.

Route	Verdict	Exact boundary or next condition
Canonical complete signed terminal	advanced	Theorem 3.1 gives a unique aggregate tower.
Three-feature to two-feature compression	advanced	Same future projection and shared trace probe required.
Continuous signed bracket	advanced	Martingale representation suffices; Clark–Ocone needs $\mathbb{D}^{1,2}$.
Representation-universal endpoint-law transport	failed	Whole projection nest must intertwine; one terminal instead uses (4.6).
R-063 as positive bracket	failed	Signed partial-Wick reconstruction has no PSD representation theorem.
Common terminal implies scalar sign	failed	Centred one-step defect (6.3) is positive.
Production collinear affine ray	advanced	Exact $P + L$ pair has curvature $2223/(25000P)$.
Global retained $P + L$ coefficient-pair convexity	failed	Active-spectator curvature (8.3) is negative.
Complete non-collinear scalar owner	open	Retain forest, future, balanced cross, low, and sextic once.

The fixed-cutoff canonical common-terminal construction itself is no longer the blocker. The most efficient next computation is the owner-complete version of the exact active-spectator direction (8.1): evaluate the polarized shared-probe trace, Doob bracket, R-063 adapted forest, future variance, balanced rows, returned low, and terminal sextic on one legal two-root canonical chart. Its mismatch is

$$\delta_o[p, q] = \bar{\tau}_o[p, q] - \mathbb{E}[\langle P_C dM_{p,o}, P_C dM_{q,o} \rangle \mid \mathcal{F}_{o-}]. \quad (9.1)$$

The aggregate must satisfy the remaining R-146 window

$$\Delta \mathcal{T}^{(0)} \leq \eta X + \zeta Y_6 + B, \quad \eta < \frac{5}{11}, \quad \zeta < \frac{13}{50}. \quad (9.2)$$

If the full companions repair (8.6), their exact mechanism becomes the next production theorem. If an owner-complete legal scaling retains an adverse direction outside (9.2), the scalar Class-II route must be reassessed before any PDE or phase claim is continued. This is a falsification test, not a preference for a spatial pattern.

10. Devil’s-advocate review

- Objection: a common-terminal tower now proves the scalar sign. UPHOLD against that inference.** The signature is indefinite, and the centred one-step fixture (6.3) has positive defect arbitrarily close to the entire trace square.
- Objection: equal terminal covariance transports the old owners. UPHOLD against that inference.** The rank-one/rank-two prefix fixture (4.5) has the same terminal law and incompatible projection nests.
- Objection: R-063 is the missing Doob bracket. UPHOLD against that inference.** Its authority is a signed finite Wick–Taylor reconstruction. Equation (5.10) has no registered positive subbracket representation.
- Objection: the generic scalar no-go condemns the actual Pauli coefficient. DISMISSED on the collinear affine class.** The exact production $P + L$ ratio has the strict curvature theorem (7.3)–(7.4).

5. **Objection: the collinear curvature proves a global production bound. UPHELD against that inference.** The exact active-spectator direction (8.3) has negative curvature. Non-collinear forest/low companions are still required.
6. **Objection: the adverse direction disproves TECT or selects a different phase. UPHELD against that inference.** The fixture omits several once-owned terms of the complete action and is not a minimizer comparison. It is only a precise route-level falsifier for global convexity of the retained $P + L$ coefficient pair.

Limit checks are explicit. A one-step filtration reduces (3.3) to ordinary polarization. A trivial auxiliary factor reduces the Bochner construction to the usual Hilbert martingale. Matching $\delta = 0$ reduces (5.7) to a negative predictable Gram. As $t \downarrow 0$, the no-go defect (6.3) vanishes; as $t \rightarrow \infty$, its ratio tends to one. At zero background the production ray remains favourable, while the non-collinear zero-floor limit is the exact negative curvature (8.5).

11. Result footer

The primary and non-importing independent executables use different finite probability filtrations and trace/bracket fixtures. Both reconstruct the production coefficients, curvature extrema, non-collinear threshold, common-terminal no-go, endpoint-law owner split, and transfer boundary. The integrated verifier reruns them, pins every authority and artefact hash, and checks the rendered PDF.

Dependencies: A1, R-063, R-075, R-104, R-125, R-130, R-136, R-141, R-142, R-146.

Exploration ledger: EXP-000615 through EXP-000622.

Registered negative and audit boundaries:

- NG-2026-08-02-A13-COMMON-TERMINAL-AUTOMATIC-SCALAR-SIGN
- NG-2026-08-02-A13-ENDPOINT-LAW-OWNER-TRANSFER
- NG-2026-08-02-A13-PRODUCTION-PAIR-GLOBAL-CONVEXITY
- AUDIT-2026-08-02-A13-R147-R063-FOREST-BRACKET-CONFLATION

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_canonical_signed_common_terminal_doob_bracket_defect_boundary_verify.py
```

Result ID: A13-CLASSII-CANONICAL-SIGNED-COMMON-TERMINAL-DOOB-BRACKET-DEFECT-BOUNDARY; Ledger ID: R-147.

Precise statement: exact canonical signed Doob terminalization and recursion; nest-intertwining and polarized trace-bracket transfer criterion; centred common-terminal and endpoint-law no-gos; strict production affine collinear P+L sign with exact non-collinear convexity counterdirection.

Scope: finite cutoff, fixed floor, actual R-141/R-142 shared feature and canonical source filtration; affine-ray theorem is collinear only.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: integrated PASS 412/412; primary 101/101; independent 93/93.

Falsification gate: any sign, factor, tower, projection-nest, bracket, curvature, threshold, hash, PDF, public-surface, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no old-owner transport, non-collinear complete scalar bound, T-050, A13, Nelson, removal, interacting measure, phase selection, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: evaluate the complete forest/future/balanced/low owner on the exact active-spectator direction and test the 5/11,13/50 window.